

ZAHIR SADIK

+8801724677163 | zahirsadik.19@gmail.com | [linkedin.com/in/zahirsadik](https://www.linkedin.com/in/zahirsadik)

Sector 14, Uttara Model Town, Dhaka-1230

OBJECTIVE

B.Sc. graduate in Electrical and Electronic Engineering from the Islamic University of Technology (IUT) with a background in power systems, renewable energy, electronics, embedded systems, robotics and machine learning. Deeply interested in teaching and mentoring, with a commitment to delivering quality education, fostering student innovation, critical thinking and practical problem solving while focusing on applying engineering knowledge and advanced technologies to real world problems and contributing to academic research.

SKILLS

- **Programming Languages:** MATLAB, Python, C, Assembly, Verilog, LaTeX
- **EDA & Simulation Software:** Proteus, PSpice, ETAP, DiGSILENT PowerFactory, PSAF, PowerWorld Simulator, Simulink, Link Planner, NYUSIM, Cadence Genus, EAGLE
- **Office & Productivity:** MS Office Suite, Microsoft Excel, Technical Documentation, Formal Report Writing.
- **Interpersonal Skills:** Leadership, Teamwork, Public Speaking, Problem solving, Presentation.

EDUCATION

- **Islamic University of Technology (IUT)** Aug 2022 – Sept 2026
Bachelor of Science in Electrical and Electronic Engineering Gazipur, Dhaka
 - CGPA: 3.86
- **Rajuk Uttara Model College** Dec 2021
Higher Secondary Certificate (HSC) Dhaka, Bangladesh
 - GPA: 5.00
- **Rajuk Uttara Model College** Feb 2019
Secondary School Certificate (SSC) Dhaka, Bangladesh
 - GPA: 5.00

HIGHLIGHTED PROJECTS

- **Multi Objective Optimization of Economic Load and Emission Dispatch in Thermal Power Generation**
Developed a multi objective optimization framework to determine optimal thermal generator scheduling by minimizing fuel cost and emissions while satisfying system power demand and generator constraints.
Tools and Techniques: Python, NumPy, SciPy, Various Optimization Methods.
- **Rooftop Solar PV System Design for a Dairy Farm in Savar, Bangladesh**
Designed a rooftop solar PV system sized to meet a dairy farm's load demand, incorporating battery sizing, MPPT selection and carbon footprint analysis.
Tools and Techniques: Solar Irradiance Data, Load Demand Analysis, PV and Battery Sizing, MPPT Selection, Carbon Footprint Analysis
- **SAP-1 Architecture-based 6-bit Computer Design and Implementation**
Designed and implemented a 6-bit computer based on the SAP-1 architecture, including digital logic design, instruction processing and hardware level simulation.
Tools and Techniques: Proteus, Digital Logic Design, Assembly Language, Hardware Simulation
- **AI Driven Renewable Forecasting and Optimal Load Dispatch for the Payra Microgrid**
Developed an AI based framework for renewable generation forecasting and optimal load dispatch in a PV microgrid using machine learning and data driven optimization techniques.
Tools and Techniques: Python, Pandas, NumPy, Machine Learning Prediction Models, Feature Engineering, Forecasting, Load Dispatch, Optimization
- **Bluetooth-Interfaced Multi-Function 8051 Embedded System**
Developed an 8051-based embedded system capable of performing multiple functions and controlled wirelessly through a Bluetooth interface.
Tools and Techniques: AT89S52, Assembly, HC-05, Proteus, LCD, Keypad

• Automated OMR Sheet Evaluation Using MATLAB Image Processing

Developed an automated OMR evaluation system in MATLAB to detect marked responses, process answer sheets and evaluate results using image processing and pixel-based analysis.

Tools and Techniques: MATLAB, Image Processing, Pixel Based Analysis, Pattern Recognition, GUI Development

TRAININGS

• Bangladesh Power Development Board (BPDB) <i>Industrial Trainee – Ghorashal Power Plant</i>	Sept 2025 Palash, Narsingdi
• Walton Hi-Tech Industries PLC <i>Industrial Trainee – Technical Division</i>	Oct 2025 Chandra, Gazipur
• Robi Axiata Limited <i>Industrial Trainee – Technical Division</i>	Nov 2025 Gulshan, Dhaka
• Bangladesh Atomic Energy Commission <i>Industrial Trainee – Electrical Division</i>	Nov 2025 Savar, Dhaka

CO-CURRICULAR AND VOLUNTEERING EXPERIENCE

• Head of Logistics <i>IEEE Robotics and Automation Society IUT Student Branch</i> ◦ Managed logistics operations for robotics competitions and society events ◦ Led planning and execution of competitions and seminars related to robotics	2023 - 2026
• Senior Electrical Subteam Member <i>Project Altair</i> ◦ Contributed to electrical system design and PCB development for robotics projects ◦ Mentored junior members in power electronics and circuit design	2023 - 2025

HIGHLIGHTED COURSES

- **Power System:** Power System I, II and III, Renewable Energy System, Power System Operation and Control, Power System Protection, Power Generation, Utilization of Electrical Energy
- **Electronics:** Electronics I & II, Digital Electronics, Semiconductor Physics, Power Electronics, VLSI Circuits, Micro-controller Based System Design
- **Communication:** Communication Engineering I & II, Wireless Communication
- **Machine Learning:** Artificial Intelligence and Machine Learning
- **Others:** Control System Engineering, Digital Signal Processing, Electromagnetic Field and Waves

UNDERGRADUATE THESIS AND RESEARCH

Uncertainty-Based Study of EV Charging Impacts Using PV, Wind and BESS in a Generic 9-Bus Microgrid with Harmonic Mitigation

This study examines harmonic distortion from EV charging in a renewable-integrated 9-bus power system, modeled in DigSILENT with PV, wind and BESS. Real world load profiles and Monte Carlo sampling capture system uncertainty and mitigation strategies are applied to maintain IEEE power quality standards.

CERTIFICATIONS AND AWARDS

• International Rover Challenge <i>Project Altair (Electrical Subteam)</i> 6th Place — ABEX Task Winner (India)	2024 Global
• Physics Brawl, International Physics Competiton <i>76th place in category O</i> Competed against 6324 competitors in 1574 teams from 81 countries	2024 Global

REFERENCES

1. Dr. Ashik Ahmed

Professor

Electrical and Electronic Engineering,
Islamic University of Technology (IUT)

Email: ashik123@iut-dhaka.edu

Phone: +880 1864-092662

2. Dr. Mohammad Tawhid Kawser

Professor

Electrical and Electronic Engineering,
Islamic University of Technology (IUT)

Email: kawser@iut-dhaka.edu

Phone: +880-1742205081